Unit 5 Guided Notes — The Pythagorean Theorem

ANSWER KEY

$a^2 + b^2 = c^2$. Square the legs and add to find c^2. To find a leg, subtract the squares.

🎯 By the end of this unit, I can…

- Use the Pythagorean Theorem to find a missing side of a right triangle
- Apply its converse to test whether a triangle is right
- Solve real-world and coordinate-distance problems

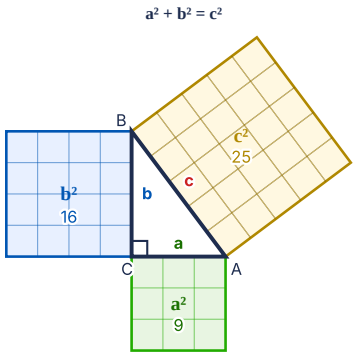
Standards: NC.8.G.6 · NC.8.G.7 · NC.8.G.8

Answer key: every blank and final answer is filled in green.

1 The Pythagorean Theorem

🔑 **THE THEOREM:** In a RIGHT triangle, the sum of the squares of the two LEGS equals the square of the HYPOTENUSE: $a^2 + b^2 = c^2$.

a, b = legs (shorter sides) · c = hypotenuse (longest side, across from the right angle)



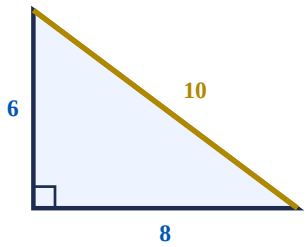
The two smaller squares (a^2+b^2) EXACTLY fill the biggest square (c^2)

⚠️ **IMPORTANT:** The Pythagorean Theorem ONLY works on RIGHT triangles (triangles with a 90° angle)!

The two shorter sides of a right triangle are called the **LEGS** (labeled a and b). The longest side, directly across from the right angle, is the **HYPOTENUSE** (labeled c). The theorem written as an equation: $a^2 + b^2 =$ **c^2**. It applies only to **RIGHT** triangles.

✏ Practice: The Pythagorean Theorem

1. ★ Try it Look at the right triangle. Which side is the hypotenuse?



- ○ 6
- ○ 8
- ● **10**

How to do it: The hypotenuse is across from the right angle (the little square). It is always the longest side. Here it is **10**.

2. ★ Try it A triangle has angles of 50°, 60°, and 70°. Can you use $a^2 + b^2 = c^2$ on it?

- ○ Yes
- ● **No**

How to do it: No angle is 90°, so it is **not** a right triangle. The theorem only works on RIGHT triangles. Answer: **No**.

3. ★★ On level The squares on the two legs of a right triangle have areas 36 and 64. What is the area of the square on the hypotenuse?

Answer: 100

How to do it: $a^2 + b^2 = c^2 \rightarrow 36 + 64 = c^2 \rightarrow c^2 =$ **100**. (So the hypotenuse is $\sqrt{100} = 10$.)

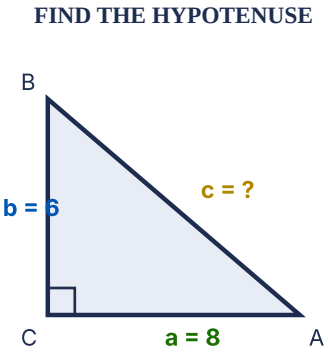
4. ★★★ Challenge The square on the hypotenuse has area 169. The square on one leg has area 25. What is the area of the square on the other leg?

Answer: 144

> **How to do it:** $a^2 + 25 = 169 \rightarrow a^2 = 169 - 25 =$ **144**. (So that leg is $\sqrt{144} = 12$.)

2 Finding the Hypotenuse

🔑 **WHEN TO USE:** When you know BOTH LEGS and need to find the HYPOTENUSE.
Formula: $c^2 = a^2 + b^2 \rightarrow c = \sqrt{(a^2 + b^2)}$



Example: Legs are 6 and 8. Find the hypotenuse.

Step 1 — Write formula: $a^2 + b^2 = c^2$

Step 2 — Substitute: $6^2 + 8^2 = c^2$

Step 3 — Square: $36 + 64 = c^2$

Step 4 — Add: $100 = c^2$

✓ **Step 5 — Square root:** $c = \sqrt{100} = 10$

Practice — Find the Hypotenuse (c)

★ Try it Legs 3 and 4: $3^2 + 4^2 = c^2$ → c = ?

c = 5

★ Try it Legs 6 and 8: $6^2 + 8^2 = c^2$ → c = ?

c = 10

★ Try it Legs 5 and 12: $5^2 + 12^2 = c^2$ → c = ?

c = 13

★ Try it Legs 8 and 15: $8^2 + 15^2 = c^2$ → c = ?

c = 17

✏ Practice: Finding the Hypotenuse

1. ★ Try it The legs of a right triangle are 12 and 16. Find the hypotenuse c.

Answer: 20

How to do it: $12^2 + 16^2 = c^2$ → 144 + 256 = c^2 → 400 = c^2 → c = √400 = **20**

2. ★★ On level The legs are 4 and 7. Find the hypotenuse c. Round to the nearest tenth (or give the exact √ answer).

Answer: c = √65 ≈ 8.1

How to do it: $4^2 + 7^2 = c^2$ → 16 + 49 = c^2 → 65 = c^2 → c = √65 ≈ 8.062 ≈ **8.1**

3. ★★★ Challenge The legs are 10 and 24. Maria says c = 34, because 10 + 24 = 34. What is the correct value of c?

Answer: 26

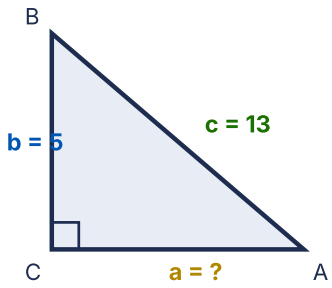
How to do it: Maria forgot to square the legs. $10^2 + 24^2 = c^2$ → 100 + 576 = 676 → c = √676 = **26**

3 Finding a Missing Leg

🔑 **WHEN TO USE:** When you know ONE LEG and the HYPOTENUSE. Formula: $a^2 = c^2 - b^2 \rightarrow a = \sqrt{(c^2 - b^2)}$

⚠️ **REMEMBER:** Finding a LEG = SUBTRACT from the hypotenuse squared. Finding the HYPOTENUSE = ADD the two legs squared.

FIND THE MISSING LEG



Example: One leg is 5, hypotenuse is 13. Find the other leg.

Step 1 — Write formula: $a^2 + b^2 = c^2$

Step 2 — Substitute: $5^2 + b^2 = 13^2$

Step 3 — Square: $25 + b^2 = 169$

Step 4 — Subtract 25 (finding a LEG = SUBTRACT): $b^2 = 144$

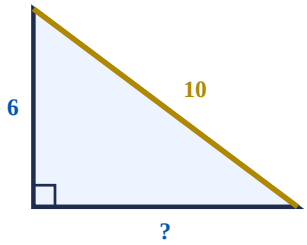
✓ **Step 5 — Square root:** $b = \sqrt{144} = 12$

ERROR ANALYSIS **A student finds a missing leg when c = 13, b = 5 by writing $a^2 = 13^2 + 5^2 = 194$. What's wrong?**

To find a leg you SUBTRACT: $a^2 = c^2 - b^2 = 169 - 25 = 144 \rightarrow a = 12$. Adding is only for the hypotenuse.

✏ Practice: Finding a Missing Leg

1. ★ Try it One leg is 6 and the hypotenuse is 10. Find the other leg.



Answer: 8

> **How to do it:** $6^2 + b^2 = 10^2 \rightarrow 36 + b^2 = 100 \rightarrow b^2 = 100 - 36 = 64 \rightarrow b = \sqrt{64} =$ **8**

2. ★★ On level One leg is 7 and the hypotenuse is 12. Find the other leg. Round to the nearest tenth (or give the exact √ answer).

Answer: $b = \sqrt{95} \approx 9.7$

> **How to do it:** $7^2 + b^2 = 12^2 \rightarrow 49 + b^2 = 144 \rightarrow b^2 = 95 \rightarrow b = \sqrt{95} \approx 9.747 \approx$ **9.7**

3. ★★★ Challenge A right triangle has a hypotenuse of 26 and one leg of 10. What is its perimeter?

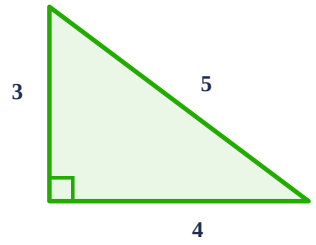
Answer: 60

> **How to do it:** Missing leg: $10^2 + b^2 = 26^2 \rightarrow 100 + b^2 = 676 \rightarrow b^2 = 576 \rightarrow b = 24$.
> Perimeter = 10 + 24 + 26 = **60**

4 Converse of the Pythagorean Theorem

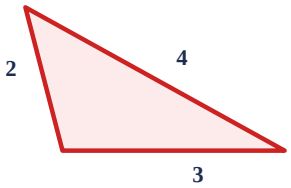
🔑 **THE CONVERSE:** Given three side lengths $a \le b \le c$, test: does $a^2 + b^2 = c^2$? If YES → right triangle. If NO → not a right triangle. Always test the two SMALLER sides against the LARGEST.

3, 4, 5
RIGHT triangle ✓



$3^2 + 4^2 = 25$ and $5^2 = 25$

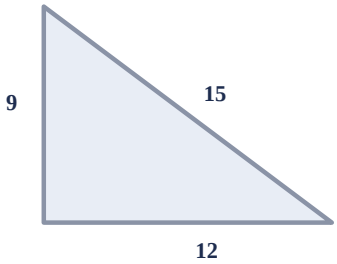
2, 3, 4
NOT a right triangle ×



$2^2 + 3^2 = 13$ but $4^2 = 16$

Example: Sides 9, 12, 15. Is it a right triangle?

9, 12, 15



Step 1 — Order the sides: $9 \le 12 \le 15$, so the LARGEST is $c = 15$

Step 2 — Write the test: does $a^2 + b^2 = c^2$?

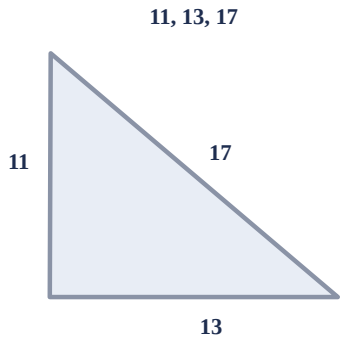
Step 3 — Substitute the two SMALLER sides: $9^2 + 12^2 =? 15^2$

Step 4 — Square each one: $81 + 144 =? 225$

Step 5 — Add the left side: $225 =? 225$

✓ Step 6 — Compare and conclude: The two sides are EQUAL, so YES — it is a right triangle.

⚠️ **Your eyes cannot do this job.** Every triangle below is drawn to true scale. Look at **11, 13, 17**: it is *not* a right triangle, because $11^2 + 13^2 = 121 + 169 =$ **290** while $17^2 =$ **289**. It misses by one. Drawn accurately it sits two-tenths of a degree off square — nobody alive could spot that. This is exactly why the converse exists: **you must run the arithmetic, every time.**



Quick Checks

★ Try it **MC** **Sides 5, 12, 13 — right triangle?**

- ● **Yes**
- ○ No

✓ $5^2 + 12^2 = 25 + 144 = 169$; $13^2 = 169$; $169 = 169$ → YES, right triangle.

★ Try it **MC** **Sides 6, 7, 10 — right triangle?**

- ○ Yes
- ● **No**

✓ $6^2 + 7^2 = 36 + 49 = 85$; $10^2 = 100$; $85 \neq 100$ → NO, not a right triangle.

✏ Practice: Converse of the Pythagorean Theorem

1. ★ Try it A triangle has sides 6, 8, and 10. Is it a right triangle?

- ● **Yes**
- ○ No

How to do it: $6^2 + 8^2 = 36 + 64 = 100$ and $10^2 = 100$. They are equal, so **Yes**.

2. ★★ On level A triangle has sides 17, 8, and 15. Is it a right triangle?

- ● **Yes**
- ○ No

> **How to do it:** Put the sides in order: 8, 15, 17. Test: $8^2 + 15^2 = 64 + 225 = 289$ and $17^2 = 289$. Equal, so **Yes**.

3. ★★ On level A triangle has sides 7, 9, and 12. Is it a right triangle?

- ○ Yes
- ● **No**

> **How to do it:** $7^2 + 9^2 = 49 + 81 = 130$, but $12^2 = 144$. $130 \neq 144$, so **No**.

4. ★★★ Challenge A triangle has sides 9, 40, and x. The side x is the longest. What must x be for a right triangle?

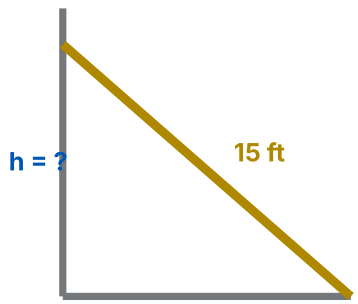
Answer: 41

> **How to do it:** For a right triangle, $9^2 + 40^2 = x^2 \rightarrow 81 + 1600 = 1681 \rightarrow x = \sqrt{1681} =$ **41**

5 Word Problems

📌 **STRATEGY:** (1) Read carefully (2) Draw & label a diagram (3) Identify legs vs hypotenuse (4) Set up $a^2 + b^2 = c^2$ (5) Solve step by step.

LADDER AGAINST A WALL



$9^2 + h^2 = 15^2$

Ladder Against a Wall: A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high does it reach?

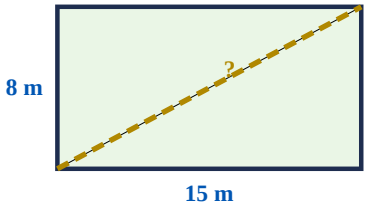
Setup: $9^2 + h^2 = 15^2$

Square & subtract: $81 + h^2 = 225 \rightarrow h^2 = 144$

✓ **Square root:** $h = \sqrt{144} = 12$ ft

✏ Practice: Word Problems

1. ★ Try it A rectangular garden is 15 m long and 8 m wide. A path goes from one corner to the opposite corner. How long is the path?



Answer: 17 m

How to do it: The path is the hypotenuse. $8^2 + 15^2 = c^2 \rightarrow 64 + 225 = 289 \rightarrow c = \sqrt{289} =$ **17 m**

2. ★★ On level A 10-foot ladder leans against a wall. The bottom is 6 feet from the wall. How high up the wall does it reach?

Answer: 8 ft

How to do it: The ladder is the hypotenuse. $6^2 + h^2 = 10^2 \rightarrow 36 + h^2 = 100 \rightarrow h^2 = 64 \rightarrow h =$ **8 ft**

3. ★★★ Challenge Sam walks 12 blocks east, then 9 blocks north. How many blocks shorter is the straight path back to where he started?

Answer: 6 blocks

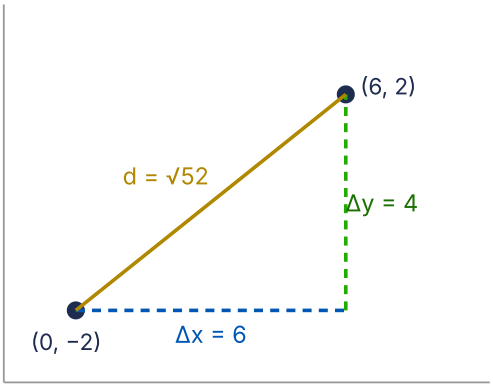
How to do it: Straight path: $12^2 + 9^2 = 144 + 81 = 225 \rightarrow \sqrt{225} = 15$ blocks. He walked $12 + 9 = 21$ blocks. $21 - 15 =$ **6 blocks** shorter.

6 Distance on the Coordinate Plane

📐 **HOW IT WORKS:** Connect the two points to form a right triangle. $\Delta x = |x_2 - x_1|$ and $\Delta y = |y_2 - y_1|$. Then use $a^2 + b^2 = c^2$ where $a = \Delta x$, $b = \Delta y$, and c = distance.

⚠️ **NO DISTANCE FORMULA NEEDED!** You already know how to do this — just use the Pythagorean Theorem!

DISTANCE = HYPOTENUSE



Example: Distance from (0, −2) to (6, 2)

Δx: $|6 - 0| = 6$

Δy: $|2 - (-2)| = 4$

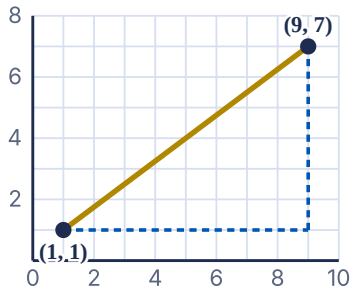
Setup: $\Delta x^2 + \Delta y^2 = d^2 \rightarrow 36 + 16 = 52$

✓ **Answer:** $d = \sqrt{52} \approx 7.21$

⚠️ **WATCH YOUR SIGNS:** When subtracting negative coordinates, subtracting a negative means ADDING! Example: 5 − (−3) = 5 + 3 = 8, NOT 5 − 3 = 2.

✏ Practice: Distance on the Coordinate Plane

1. ★ Try it Find the distance between (1, 1) and (9, 7).



Answer: 10

How to do it: $\Delta x = 9 - 1 = 8$ and $\Delta y = 7 - 1 = 6$. $8^2 + 6^2 = 64 + 36 = 100 \rightarrow d = \sqrt{100} =$ **10**

2. ★★ On level Find the distance between (−2, 3) and (3, −1). Round to the nearest tenth (or give the exact √ answer).

Answer: $d = \sqrt{41} \approx 6.4$

How to do it: $\Delta x = |3 - (-2)| = 5$ and $\Delta y = |-1 - 3| = 4$. $5^2 + 4^2 = 25 + 16 = 41 \rightarrow d = \sqrt{41} \approx 6.403 \approx$ **6.4**

3. ★★★ Challenge Point P is in Quadrant I. It is 13 units from (0, 0), and its x-coordinate is 5. What is its y-coordinate?

Answer: 12

How to do it: The distance 13 is the hypotenuse and 5 is one leg. $5^2 + y^2 = 13^2 \rightarrow 25 + y^2 = 169 \rightarrow y^2 = 144 \rightarrow y =$ **12**. So P = (5, 12).

✏ Part 2 — Comprehensive Practice

📌 **Show every step on its own line.** Box your final answer.

✏ Comprehensive Practice All sections

1. ★ Try it A right triangle has legs 5 and 12 and hypotenuse 13. Which equation is true?

- ● **$5^2 + 12^2 = 13^2$**
- ○ $5^2 + 13^2 = 12^2$
- ○ $5 + 12 = 13$

> **How to do it:** The two legs go on the left. The hypotenuse goes alone on the right: $5^2 + 12^2 = 13^2$ (25 + 144 = 169). **$5^2 + 12^2 = 13^2$**

2. ★ Try it The legs of a right triangle are 15 and 20. Find the hypotenuse.

Answer: 25

> **How to do it:** $15^2 + 20^2 = c^2 \rightarrow 225 + 400 = 625 \rightarrow c = \sqrt{625} =$ **25**

3. ★ Try it A triangle has sides 12, 16, and 20. Is it a right triangle?

- ● **Yes**
- ○ No

> **How to do it:** $12^2 + 16^2 = 144 + 256 = 400$ and $20^2 = 400$. Equal, so **Yes**.

4. ★★ On level One leg is 9 and the hypotenuse is 41. Find the other leg.

Answer: 40

> **How to do it:** $9^2 + b^2 = 41^2 \rightarrow 81 + b^2 = 1681 \rightarrow b^2 = 1600 \rightarrow b = \sqrt{1600} =$ **40**

5. ★★ On level The legs are 2 and 5. Find the hypotenuse. Round to the nearest tenth (or give the exact √ answer).

Answer: **c = √29 ≈ 5.4**

> **How to do it:** $2^2 + 5^2 = c^2 \rightarrow 4 + 25 = 29 \rightarrow c = \sqrt{29} \approx 5.385 \approx$ **5.4**

6. ★★ On level A triangle has sides 10, 24, and 25. Is it a right triangle?

○ Yes

● **No**

> **How to do it:** $10^2 + 24^2 = 100 + 576 = 676$, but $25^2 = 625$. $676 \neq 625$, so **No**.

7. ★★ On level A boat sails 7 miles east, then 24 miles north. How far is it from where it started?

Answer: **25 miles**

> **How to do it:** The straight-line distance is the hypotenuse. $7^2 + 24^2 = 49 + 576 = 625 \rightarrow \sqrt{625} =$ **25 miles**

8. ★★ On level Find the distance between (−3, 2) and (3, 10).

Answer: **10**

> **How to do it:** $\Delta x = |3 - (-3)| = 6$ and $\Delta y = |10 - 2| = 8$. $6^2 + 8^2 = 36 + 64 = 100 \rightarrow d = \sqrt{100} =$ **10**

9. ★★★ Challenge A square has a diagonal of 10 cm. How long is each side? Round to the nearest tenth (or give the exact √ answer).

Answer: **s = √50 ≈ 7.1 cm**

> **How to do it:** Both legs are the same length, s. $s^2 + s^2 = 10^2 \rightarrow 2s^2 = 100 \rightarrow s^2 = 50 \rightarrow s = \sqrt{50} \approx 7.071 \approx$ **7.1 cm**

10. ★★★ Challenge A triangle has corners at (0, 0), (4, 0), and (4, 3). What is its perimeter?

Answer: **12**

> **How to do it:** Bottom side: 4. Right side: 3. Slanted side: $4^2 + 3^2 = 16 + 9 = 25 \rightarrow \sqrt{25} = 5$. Perimeter = 4 + 3 + 5 = **12**

11. ★★★ Challenge Explain how you know when to ADD the squares and when to SUBTRACT them.

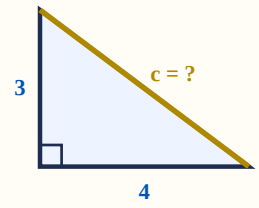
Answer: Find the hypotenuse → add. Find a leg → subtract.

> **How to do it:** If you know both legs and want the hypotenuse, ADD: $c^2 = a^2 + b^2$. If you know the hypotenuse and one leg, SUBTRACT: $a^2 = c^2 - b^2$.

More Practice by Skill

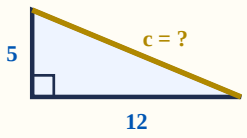
A. Find the Hypotenuse

★ Try it 1) Legs 3, 4. c = ?



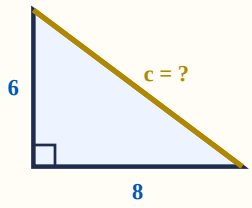
c = 5

★ Try it 2) Legs 5, 12. c = ?



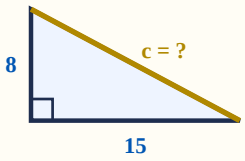
c = 13

★ Try it 3) Legs 6, 8. c = ?



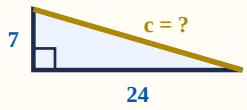
c = 10

★ Try it 4) Legs 8, 15. c = ?



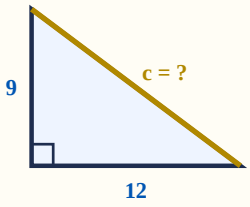
c = 17

★★ On level 5) Legs 7, 24. c = ?



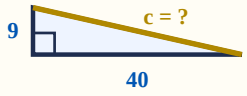
c = 25

★★ On level 6) Legs 9, 12. c = ?



c = 15

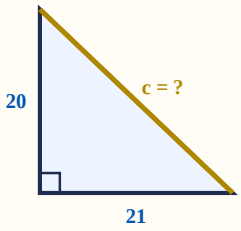
★★ On level 7) Legs 9, 40. c = ?



c =

41

★★ On level 8) Legs 20, 21. c = ?

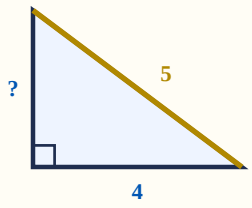


c =

29

B. Find the Missing Leg

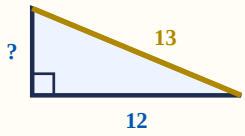
★ Try it 1) Known leg 4, hyp 5. Leg = ?



leg =

3

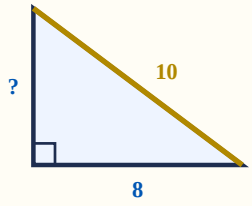
★ Try it 2) Known leg 12, hyp 13. Leg = ?



leg =

5

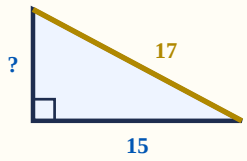
★ Try it 3) Known leg 8, hyp 10. Leg = ?



leg =

6

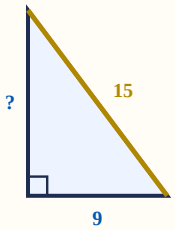
★ Try it 4) Known leg 15, hyp 17. Leg = ?



leg =

8

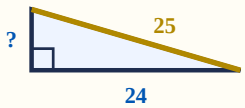
★★ On level 5) Known leg 9, hyp 15. Leg = ?



leg =

12

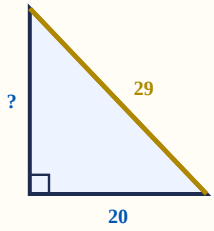
★★ On level 6) Known leg 24, hyp 25. Leg = ?



leg =

7

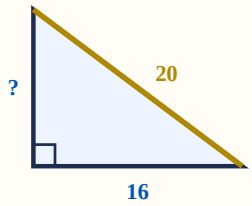
★★ On level 7) Known leg 20, hyp 29. Leg = ?



leg =

21

★★ On level 8) Known leg 16, hyp 20. Leg = ?

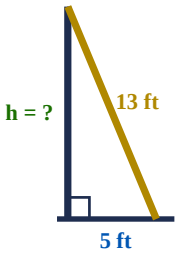


leg =

12

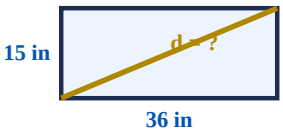
C. Word Problems

★★★ Challenge **1.** A 13-foot ladder leans against a wall. The base is 5 feet from the wall. How high does it reach?



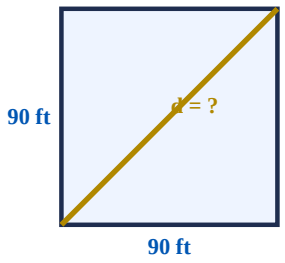
$h^2 + 5^2 = 13^2 \rightarrow h^2 = 169 - 25 = 144 \rightarrow h = 12$ ft.

★★ On level **2.** A rectangular TV is 36 inches wide and 15 inches tall. What is the diagonal length?



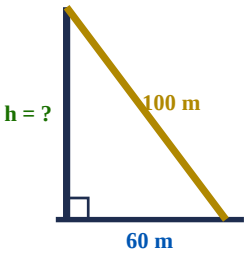
$d^2 = 36^2 + 15^2 = 1296 + 225 = 1521 \rightarrow d = 39$ in.

★★★ Challenge **3.** A baseball diamond is a square with 90-ft sides. Distance from home to second base?



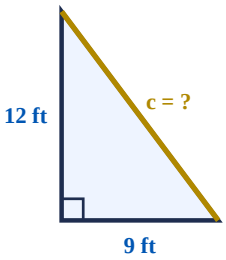
d² = 90² + 90² = 16200 → d ≈ 127.3 ft.

★★★ Challenge **4.** A kite string is 100 m long. The kite is directly above a point 60 m away. How high is the kite?



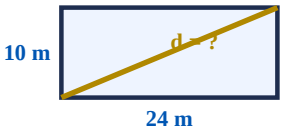
h² + 60² = 100² → h² = 10000 − 3600 = 6400 → h = 80 m.

★★ On level **5.** A support beam covers 9 ft horizontal and 12 ft vertical. How long is the beam?



c² = 9² + 12² = 81 + 144 = 225 → c = 15 ft.

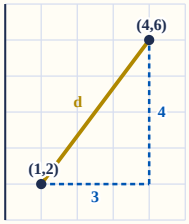
★★ On level **6.** A pool is 24 m long and 10 m wide. A swimmer swims diagonally. How far?



d² = 24² + 10² = 576 + 100 = 676 → d = 26 m.

D. Distance on the Coordinate Plane

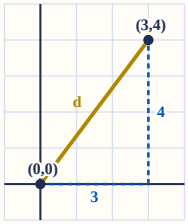
★ Try it 1) (1,2) → (4,6). d = ?



d =

5

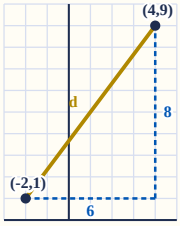
★ Try it 2) (0,0) → (3,4). d = ?



d =

5

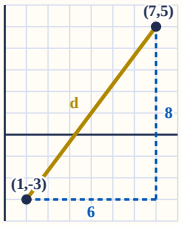
★★ On level 3) (−2,1) → (4,9). d = ?



d =

10

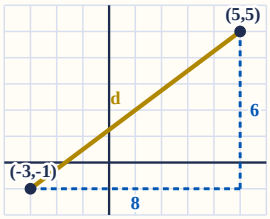
★★ On level 4) (1,−3) → (7,5). d = ?



d =

10

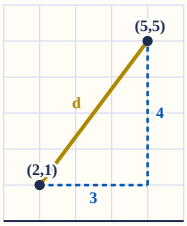
★★ On level 5) (−3,−1) → (5,5). d = ?



d =

10

★ Try it 6) (2,1) → (5,5). d = ?



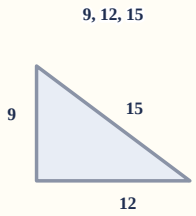
d =

5

E. The Converse — Is It a Right Triangle?

Test the two **smaller** sides against the **largest**. Each figure is drawn to true scale, so do not trust the picture — two of these miss by a hair.

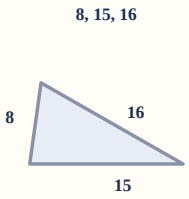
★ Try it 1) Sides 9, 12, 15 — right triangle? (write *yes* or *no*)



Answer:

Yes — $9^2 + 12^2 = 225$, and $15^2 = 225$, so $225 = 225$

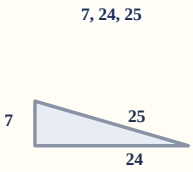
★★ On level 2) Sides 8, 15, 16 — right triangle? (write *yes* or *no*)



Answer:

No — $8^2 + 15^2 = 289$, and $16^2 = 256$, so $289 \neq 256$

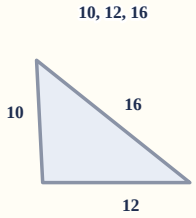
★ Try it 3) Sides 7, 24, 25 — right triangle? (write *yes* or *no*)



Answer:

Yes — $7^2 + 24^2 = 625$, and $25^2 = 625$, so $625 = 625$

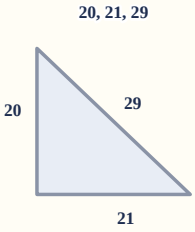
★★ On level 4) Sides 10, 12, 16 — right triangle? (write *yes* or *no*)



Answer:

No — $10^2 + 12^2 = 244$, and $16^2 = 256$, so $244 \neq 256$

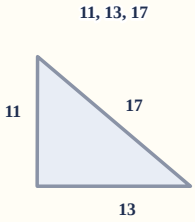
★★★ Challenge 5) Sides 20, 21, 29 — right triangle? (write *yes* or *no*)



Answer:

Yes — $20^2 + 21^2 = 841$, and $29^2 = 841$, so $841 = 841$

★★★ Challenge 6) Sides 11, 13, 17 — right triangle? (write *yes* or *no*)



Answer:

No — $11^2 + 13^2 = 290$, and $17^2 = 289$, so $290 \neq 289$